

Disproof of conjecture 00000001186

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Verdict: FALSE

We disprove conjecture 00000001186 as stated. The disproof consists of three independent internal contradictions. It is unconditional: it does not rely on the classification of finite simple groups, nor on any structural theory of finite groups.

1 The conjecture

Definition 1 (as stated). Let $m(k)$ denote the minimal order of a nonabelian simple group with exactly k distinct prime factors.

Conjecture. $m(3) = 60$ (A_5), $m(4) = 504$ ($\text{PSL}_2(8)$), $m(5) = 660$ ($\text{PSL}_2(11)$); and $m(k)/m(k-1) \leq 4$ for all k , with equality at $k = 4$.

Throughout, $\omega(n)$ denotes the number of *distinct* primes dividing n , and $\Omega(n)$ denotes the number of primes dividing n counted *with multiplicity*. The definition of “exactly k distinct prime factors” for a group G means $\omega(|G|) = k$: a group with exactly k distinct prime factors is one whose order is divisible by exactly k distinct primes.

2 The three contradictions

Contradiction 1: 504 has three distinct prime factors, not four

Lemma 2. $504 = 2^3 \cdot 3^2 \cdot 7$, hence $\omega(504) = 3$.

Proof. Direct computation: $2^3 = 8$, $3^2 = 9$, and $8 \cdot 9 \cdot 7 = 72 \cdot 7 = 504$. The prime divisors are 2, 3, and 7, which are pairwise distinct, so $\omega(504) = 3$. $\square$

Theorem 3. *No group of order 504 has exactly four distinct prime factors. Consequently, on the definition above, $m(4) \neq 504$.*

Proof. Let G be a group with $|G| = 504$. The distinct prime factors of G (as the definition uses the phrase) are the distinct primes dividing $|G| = 504$. By Lemma 1 these are exactly 2, 3, 7, so $\omega(|G|) = 3$. In particular $\omega(|G|) \neq 4$: no group of order 504 has exactly four distinct prime factors. Since $m(4)$ is by definition the order of such a group, the value 504 cannot equal $m(4)$. $\square$

Remark 4. This contradiction is unconditional. It does not require knowing whether a nonabelian simple group of order 504 exists, nor which one is intended: for *every* group of order 504, the order has only three distinct prime divisors, whereas the definition of $m(4)$ demands exactly four. Clause (1) above stands on its own.

Contradiction 2: 660 has four distinct prime factors, not five

Lemma 5. $660 = 2^2 \cdot 3 \cdot 5 \cdot 11$, hence $\omega(660) = 4$.

Proof. $2^2 = 4$ and $4 \cdot 3 \cdot 5 \cdot 11 = 12 \cdot 55 = 660$. The prime divisors 2, 3, 5, 11 are pairwise distinct, so $\omega(660) = 4$. $\square$

Theorem 6. No group of order 660 has exactly five distinct prime factors. Consequently, on the definition above, $m(5) \neq 660$.

Proof. If $|G| = 660$, then the distinct prime factors of G are the distinct primes dividing 660, namely 2, 3, 5, 11; thus $\omega(|G|) = 4 \neq 5$. Hence no group of order 660 has exactly five distinct prime factors, and $m(5) \neq 660$. Equivalently, 660 is a candidate value for $m(4)$, not for $m(5)$: the statement has mislabelled it by one index. $\square$

Contradiction 3: the ratio bound fails on the conjecture's own numbers

Theorem 7. Using the values listed in the conjecture,

$$\frac{m(4)}{m(3)} = \frac{504}{60} = \frac{42}{5} = 8.4 > 4.$$

This contradicts the asserted inequality $m(k)/m(k-1) \leq 4$ for all k , and in particular contradicts the assertion that equality holds at $k = 4$.

Proof. Reduce the fraction: $504/60 = (504/12)/(60/12) = 42/5$. Since $42/5 = 8.4$ and $8.4 > 4$, the ratio exceeds 4. The conjecture asserts $m(k)/m(k-1) \leq 4$ for all k ; taking $k = 4$ and substituting the stated values $m(4) = 504$, $m(3) = 60$ gives $8.4 \leq 4$, a false numerical statement. $\square$

Remark 8. Contradiction 3 is pure arithmetic on the numbers supplied by the conjecture itself. It is independent of what those numbers are taken to name and of the meaning assigned to “prime factors”.

3 Factorisation table

n	factorisation	$\omega(n)$ (distinct)	$\Omega(n)$ (multiplicity)
60	$2^2 \cdot 3 \cdot 5$	3	4
504	$2^3 \cdot 3^2 \cdot 7$	3	6
660	$2^2 \cdot 3 \cdot 5 \cdot 11$	4	5

Thus $\omega(504) = 3 \neq 4$ (Contradiction 1), $\omega(660) = 4 \neq 5$ (Contradiction 2), and $504/60 = 8.4 > 4$ (Contradiction 3). Each of the three contradictions alone falsifies the conjecture; they are independent in the sense that removing any one still leaves the conjecture false by the other two.

4 The multiplicity reading rescues nothing

A reader might attempt to read “exactly k distinct prime factors” as “ $\Omega(|G|) = k$ ”, i.e. counting prime factors with multiplicity. This does not save the conjecture. From the table:

- $\Omega(60) = 2 + 1 + 1 = 4 \neq 3$. Under the multiplicity reading the very first listed value, $m(3) = 60$, already fails.
- $\Omega(504) = 3 + 2 + 1 = 6 \neq 4$, so $m(4) = 504$ still fails.
- $\Omega(660) = 2 + 1 + 1 + 1 = 5$. This is the only listed number that becomes consistent under the multiplicity reading, but it is assigned to $m(5)$; it would make 660 a candidate for $m(5)$, not for $m(4)$.

Moreover the ratio bound is a statement about the listed numbers themselves: $504/60 = 8.4 > 4$ holds under either reading. Therefore no reading of “prime factors” makes all of the listed values simultaneously correct, and the conjecture is false as stated.

5 True values on the standard reading (context only)

The disproof above does not need the following; we record it as context. On the standard reading, using ω , the initial values are

$$m(3) = 60 \text{ (A}_5\text{)}, \quad m(4) = 660 \text{ (PSL}_2(11)\text{)}.$$

Indeed $660 = 2^2 \cdot 3 \cdot 5 \cdot 11$ has exactly four distinct prime divisors, and every nonabelian simple group of order strictly less than 660 has at most three distinct prime divisors. The orders occurring below 660 are

$$60 = 2^2 \cdot 3 \cdot 5, \quad 168 = 2^3 \cdot 3 \cdot 7, \quad 360 = 2^3 \cdot 3^2 \cdot 5, \quad 504 = 2^3 \cdot 3^2 \cdot 7,$$

each with exactly three distinct prime divisors; here $|\text{A}_5| = 60$, $|\text{PSL}_2(7)| = 168$, $|\text{A}_6| = 360$, while 504 is not the order of any nonabelian simple group. Hence, on the standard reading, the statement’s “ $m(4) = 504$ ” is wrong, and its “ $m(5) = 660$ ” is a one-index mislabelling of the correct $m(4)$.

We stress that this section is context only. The refutation rests solely on the three contradictions, of which Contradictions 1 and 2 require only the factorisations $504 = 2^3 \cdot 3^2 \cdot 7$ and $660 = 2^2 \cdot 3 \cdot 5 \cdot 11$, and Contradiction 3 requires only $504/60 = 8.4 > 4$. In particular the refutation does not depend on classifying simple groups.